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**Course Code**: CSA0903

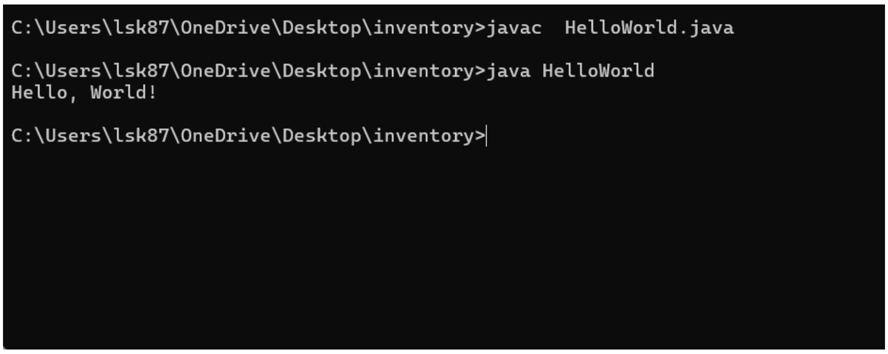
**Course Name**: Programming in JAVA

## Post Class Assignment 1

1. Write five advantages of Java and compare Java with C/C++.

- **Platform Independent** – Java programs run on any operating system using the Java Virtual Machine (JVM).
- **Object-Oriented** – Supports concepts like classes, objects, inheritance, and polymorphism.
- **Secure** – Provides features like bytecode verification, no direct pointer access, and automatic memory management.
- **Robust** – Has strong exception handling and automatic garbage collection, reducing crashes and memory leaks.
- **Multithreaded** – Allows multiple tasks to run simultaneously, improving application performance

2. Install Java successfully and submit a screenshot along with the output of the Hello World program.

3. Write programs demonstrating variable scope and lifetime.

```
class ScopeDemo {
    static int staticVar = 100;
    int instanceVar = 50;

    public static void main(String[] args) {
        ScopeDemo obj = new ScopeDemo();

        int localVar = 10;

        System.out.println("Local Variable: " + localVar);
        System.out.println("Instance Variable: " +
obj.instanceVar);
        System.out.println("Static Variable: " + staticVar);

        {
            int blockVar = 20;
            System.out.println("Block Variable: " + blockVar);
        }
    }
}
```

4. Write programs for array sum, average, reverse, and second-largest element.

## Array Sum

```
class ArraySum {
    public static void main(String[] args) {
        int[] a = {10, 20, 30, 40, 50};
        int sum = 0;

        for (int i = 0; i < a.length; i++) {
            sum += a[i];
        }

        System.out.println("Sum = " + sum);
    }
}
```

## Array Average

```
class ArrayAverage {
    public static void main(String[] args) {
        int[] a = {10, 20, 30, 40, 50};
        int sum = 0;

        for (int i = 0; i < a.length; i++) {
            sum += a[i];
        }

        double avg = (double) sum / a.length;
        System.out.println("Average = " + avg);
    }
}
```

## Reverse Array

```
class ReverseArray {
    public static void main(String[] args) {
        int[] a = {10, 20, 30, 40, 50};

        for (int i = a.length - 1; i >= 0; i--) {
            System.out.print(a[i] + " ");
        }
    }
}
```

## Second Largest Element

```
class SecondLargest {
    public static void main(String[] args) {
        int[] a = {10, 50, 30, 40, 20};

        int largest = a[0];
        int second = a[0];

        for (int i = 1; i < a.length; i++) {
            if (a[i] > largest) {
                second = largest;
                largest = a[i];
            } else if (a[i] > second && a[i] != largest) {
                second = a[i];
            }
        }

        System.out.println("Second Largest = " + second);
    }
}
```